Clinical Decision Making and Pattern Recognition in Health Care

Agentic Generative AI, Classification, Prediction, and Anomaly Detection for Treatment, Payment & Operations

Concept Definition

The field of decision making and pattern recognition in health care involves the intelligent use of artificial intelligence to recognize, categorize, predict, and take action based on patterns contained within clinical, administrative, and financial information. Essentially, this field is concerned with solving an age-old problem in medicine -- that of converting diverse streams of data into useful decision making processes.

Modern approaches use a number of technologies that converge in some way.

Classification models differentiate between diagnoses, procedure codes, and fraud indicators at a claim level. Predictive models predict patient decline, risks for readmission, and high-cost utilization trends. Time-series anomaly detection uncovers irregularities in billing patterns, treatment frequency, and operational throughput that point to fraud, waste, or patient deterioration. Chain reasoning is a process where a model breaks down a complicated medical question into a sequence of logical steps.

Most notably, the agentic generative AI defines the present-day horizon. This kind of system does not simply classify information or create outputs based on its analysis but rather independently breaks down multi-stage goals, coordinates specialists in separate tasks and changes strategy in accordance with results. In the healthcare sector, the agentic approach could be used to search through EHRs, compare to billing policy, follow guidelines, detect anomalies, and provide an auditable recommendation—within one process flow. It means that the focus will move from assistants to TPO co-managers.

Trends

Several intersecting trends define this landscape:

- Acceleration of agentic deployment: Gartner projects that agentic AI, representing less than 1 percent of enterprise software in 2024, will expand rapidly through 2028 as healthcare organizations move beyond pilots. Leading health systems such as MUSC Health and Mayo Clinic have initiated agentic AI programs targeting back-office automation and clinical workflow coordination.
- Move from retrospective to prepayment analytics: AI-powered payment integrity is evolving from post-payment audit to real-time prepayment review. Machine learning enhances the accuracy of rules engines, cutting down false-positive audit load while detecting more mistakes prior to payments.
- Multiple agent orchestration in clinical environments: Architectures used for research like DynamiCare prove multiple agent systems that can engage in interactive and open-ended medical decision-making, where agents focus on history-taking, differential diagnosis, guidelines search, and therapy selection.
- Regulatory maturity: More than 1,250 AI-powered medical devices were authorized by the FDA by late 2025; however, fully autonomous clinical agents are subject to strict Class III regulatory pathway. Regulatory frameworks are currently developing to address AI-based decision support tools without compromising physician liability.
- Explainability as a mandatory feature: Payors, providers, and regulators require reasoning chains that are auditable. Chain-of-thought and structured inference approaches become obligatory features rather than nice-to-have additions.

Opportunities and Threats

There are many opportunities for organizations at the confluence of healthcare data and payment integrity solutions. Agentic AI can help save on administrative costs, which amount to 20% of organizational expenditures, by automating prior authorization, coding verification, and claim review processes. Anomaly detection algorithms trained on historical claims data will be able to detect instances of fraud, waste, and abuse that may escape rule-based methods, especially those using coding loopholes. Machine learning models can be used to find high-risk patients ahead of time, allowing for targeted interventions that will lower cost of care. Combining the SDOH data with clinical and claims data allows for better feature sets.

Also, threats are just as important. The hallucination issue of large language models is associated with a risk of harm to patients since recommendations made by AI are not reviewed properly in the context of clinical and coding advice. Discriminatory patterns related to demographic and historical biases from training data will continue to be reflected in the authorization of treatments and risk scoring. An adversarial party may use machine learning algorithms that can identify undetected billing patterns, thereby outplaying anomaly detection tools. Liability concerns related to autonomy also represent a risk of adoption. Lastly, interoperability issues affect the diversity of training data.

Strategic Recommendations for Cotiviti

Given Cotiviti’s established position as the highest-designated leader in payment integrity solutions and its formalized AI Center of Excellence, three strategic investments merit priority consideration:

- Use Agentic Prepayment Reasoning: Create an agential layer that combines Cotiviti’s current rules on payment accuracy with the reasoning capabilities of LLMs in real time. The use of an agent specialized in policy discovery along with one specialized in classifying billing irregularities and another in chain reasoning will help make prepayment determinations with better accuracy than what can currently be achieved by their rules-based system.
- Time Series Anomaly Detection for TPO Insights: Invest in capabilities that will allow for tracking the longitudinal claims stream -- not just single claims -- looking for any anomalies in the treatment and billing processes. The provider level time series analysis will allow for identification of upcoding and unbundling.
- Establish an Explainability and Audit Framework: Differentiate Cotiviti’s AI offerings by embedding structured audit trails - capturing the reasoning chain behind every AI-generated recommendation - into its analytics platform. As payers face regulatory scrutiny of AI-driven claim denials, an auditable and defensible AI layer becomes a competitive differentiator and a risk mitigation tool.

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